

FLUXPAY

M1 Backend Testing Report

Authentication, user profile, KYC, administration,
security, and Oracle verification

TESTED - PASS

Report status	Completed - all executed automated tests passed
Test execution date	11 September 2026
Application under test	FluxPay M1 backend (Spring Boot, Oracle, JWT)
Test approaches	Postman API checks, unit tests, MockMvc controller tests, Oracle integration test
Automated result	42 tests run; 0 failures; 0 errors; 0 skipped

Purpose: This document records what was tested, expected and observed behaviour, representative test data, and defects identified and resolved during the M1 backend test phase.

1. Executive summary

The FluxPay M1 backend was tested successfully using manual API verification in Postman and a repeatable automated test suite. The verified scope covers authentication, user self-service, KYC submission and lifecycle handling, administrator review, JWT security responses, wallet provisioning integration, and Oracle persistence.

The backend is suitable to move forward from the backend test phase. The dedicated Oracle test schema remains available for future repeatable integration runs.

Test outcome

Area	Result	Evidence
Backend startup and database	PASS	Oracle connection established; Flyway applied migrations through V502.
Manual API testing	PASS	Nine endpoints and negative/security flows exercised with Postman.
Automated test suite	PASS	42 passed; 0 failed; 0 errors; 0 skipped.
Oracle integration test	PASS	Dedicated FLUXPAY_TEST schema; migration, API flow, repository verification, cleanup.
Open backend blockers	NONE	Issues found during testing were resolved and verified.

Environment and configuration

Component	Configuration used
Backend profile	local,m1-solo
Application URL	http://localhost:8080
Database	Oracle Free PDB (FREEPDB1)
Main application schema	FLUXPAY
Automated integration schema	FLUXPAY_TEST (dedicated test-only user)
Wallet integration	M1 mock wallet provisioner, active only under m1-solo
API contract check	OpenAPI available at GET /v3/api-docs

2. Manual API testing - Postman

The following checks were executed against a running local backend with Oracle. Authentication tokens and passwords are intentionally excluded from this report.

Endpoint / scenario	Expected result	Observed result
GET /v3/api-docs	OpenAPI document returned	PASS - OpenAPI 3.0.1 returned.
POST /api/auth/register	201; USER role; KYC NONE; JWT envelope	PASS - registered test USER with KYC NONE.
Duplicate canonical registration	409 EMAIL_EXISTS	PASS - padded/mixed-case duplicate rejected.
POST /api/auth/login	200; one-hour JWT	PASS - Bearer token, expiresIn 3600.
Invalid credentials	401 INVALID_CREDENTIALS; no account disclosure	PASS - correct generic response.
GET /api/users/me	200 current user only	PASS - profile and derived KYC status returned.
PUT /api/users/me	200; fullName change only	PASS - name updated; protected fields unchanged.
Missing / invalid JWT	401 AUTH_REQUIRED	PASS - JSON security envelope returned.
Padded/mixed-case legacy login	Canonicalized login succeeds	PASS - login returned 200 after correction.

KYC user API checks

Scenario	Representative test data	Observed result
Initial status	New user with no KYC case	PASS - 200; NONE; id/version null.
Empty submission	No document metadata	PASS - 400 VALIDATION.
Valid submission	PAN ABCDE1234F; pan-card.pdf; application/pdf; 102400 bytes	PASS - 201; PENDING; version 0.
Pending re-submit	Second valid request while PENDING	PASS - 409 KYC_ALREADY_PENDING.
Unsupported type	application/octet-stream	PASS - 400 VALIDATION.
Oversized file	File size over 5 MiB	PASS - 400 VALIDATION.
Verified re-submit	Submission after approval	PASS - 409 KYC_ALREADY_VERIFIED.

3. Admin KYC review and state transition testing

Admin operations were tested using a separately authenticated ADMIN account. A USER token was deliberately used against the admin list endpoint to verify role enforcement.

Scenario	Expected	Observed result
USER calls admin list	403 FORBIDDEN	PASS - USER-to-admin denial verified.
ADMIN lists PENDING applications	200 with pending KYC row	PASS - KYC metadata, document, status and version returned.
Reject without reason	400 REJECT_REASON_REQUIRED	PASS - blank reason rejected.
Reject with reason	200; REJECTED; version increment	PASS - reason saved; version 0 to 1.
Rejected user re-submits	201; PENDING; decision reset	PASS - version 1 to 2; reason/time cleared.
Stale approve	409 KYC_CONFLICT	PASS - outdated expectedVersion rejected.
Approve current PENDING case	200; VERIFIED; audit data updated	PASS - version 2 to 3; VERIFIED.
User status after approval	Profile and KYC status VERIFIED	PASS - both endpoints confirmed VERIFIED.

Verified lifecycle evidence

Stage	Status	Version	Important observed data
No application	NONE	null	No application id, submission time, or decision time.
First submission	PENDING	0	PAN document metadata accepted.
Admin rejection	REJECTED	1	Reason: Document image needs to be clearer.
Resubmission	PENDING	2	Rejection fields reset.
Admin approval	VERIFIED	3	Decision timestamp populated; profile status derived correctly.

4. Automated test evidence

Automated tests were executed with Maven against the backend project. Complete suite result: **Tests run: 42, Failures: 0, Errors: 0, Skipped: 0; BUILD SUCCESS.**

Test class / layer	Tests	Coverage verified
JwtUtilTest	3	Valid signed JWT, tampered token, expired token.
M1ControllerMvcTest	16	All nine endpoints; success paths; authorization and key error contracts.
OracleM1ApiIntegrationTest	1	Real Oracle migration, registration, JWT-authenticated profile, KYC persistence/read and cleanup.
AuthServiceTest	6	Canonical registration, BCrypt, wallet provision, duplicate handling, padded login, credentials, KYC state.
DatabaseKycGateTest	4	Gate true only for VERIFIED; all other states denied.
KycServiceTest	8	Validation and NONE/PENDING/REJECTED/VERIFIED lifecycle and conflicts.
UserServiceTest	4	Current-user reads, permitted name update, KYC-status derivation.
Total	42	ALL PASSED

Oracle integration verification

A separate FLUXPAY_TEST schema was used to avoid altering application data. Flyway applied the full migration history through V502. The integration test created a unique user and KYC application through the HTTP layer, verified persistence via repositories, then removed transient data after execution.

Repeat command: Set ORACLE_TEST_JDBC_URL, ORACLE_TEST_USERNAME and ORACLE_TEST_PASSWORD for the dedicated test schema, then run `.\mvnw.cmd -f backend\pom.xml test`. The backend server does not need to be running.

5. Issues found and resolved

The following issues were identified during test preparation or execution. Each was corrected and retested.

ID	Issue found	Resolution	Verification
T-01	Padded/mixed-case email login failed request validation before canonicalization.	Removed strict email-format validation from LoginRequest; retained required/length validation so AuthService can trim and lowercase legacy input.	Postman padded/mixed-case login returned 200.
T-02	Authenticated admin requests initially returned AUTH_REQUIRED while testing rejection.	Corrected request route to /api/admin/kyc/applications/{id}/reject. JWT filter registration was also made explicit in the security chain.	Blank reason returned 400; valid reject returned 200.
T-03	Oracle test migration failed because test schema lacked identity/sequence privileges.	Recreated FLUXPAY_TEST with quota and CREATE SESSION, TABLE, TRIGGER, and SEQUENCE privileges.	Flyway to V502 and Oracle integration test passed.
T-04	Initial Oracle privilege script included CREATE INDEX, which was invalid in this grant context.	Removed invalid privilege; granted only required valid privileges.	Test-user setup completed successfully.

Known non-blocking observation

Flyway emitted a compatibility warning because local Oracle is newer than the latest explicitly supported version in the bundled Flyway release. It did not prevent migrations or tests: V502 and all integration checks completed successfully. Review this during dependency maintenance; it is not a current test failure.

6. Conclusion and handover

The M1 backend passed its defined manual and automated test scope. Security behavior, role authorization, KYC state controls, conflict handling, data validation, wallet provisioning rollback coverage, and Oracle persistence were all demonstrated. The backend is ready for the next project phase, with FLUXPAY_TEST retained for repeatable integration testing.

BACKEND TEST PHASE COMPLETE